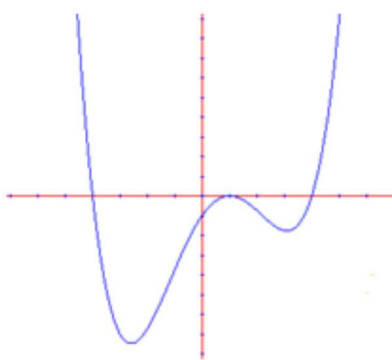


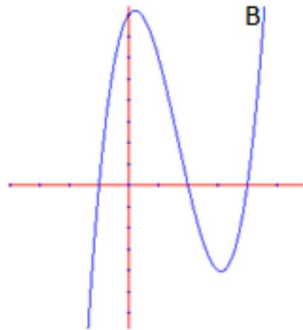
Unit 1 Review – Polynomial Functions

1. Fill in the blanks.

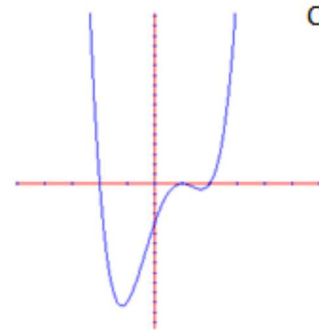
- a) State the remainder when $-4x^3 + 3x^2 + 2x - 1$ is divided by $x - 2$. $f(2) = -17$
- b) State the roots and the order of each root of $g(x) = 2x^2(2x + 3)^3$. 0 (order 2), $-\frac{3}{2}$ (order 3)
- c) When a function is divided by $2x - 1$, the remainder is -2 ; Determine the remainder when the same function is divided by $x - \frac{1}{2}$. -2
- d) Values that could be zeros for the polynomial $f(x) = 4x^3 + 2x^2 - 7x - 8$ are $\pm\{1, 2, 4, 8, \frac{1}{2}, \frac{1}{4}\}$
- e) State if $y = -2x^4 + 3x^2 + 1$ is odd, even or neither. even
- f) Beside each equation below, put the letter of the graph that best describes the equation:
- i) $y = (x^2 - 16)(x - 1)^2$ A
- ii) $y = (2 - x)(x - 4)(x + 1)$ E



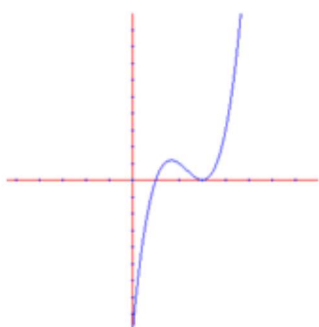
A



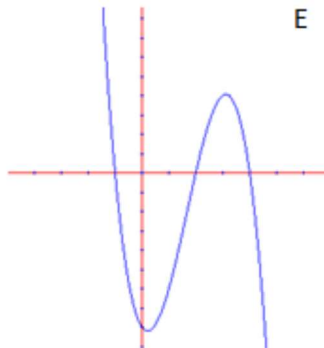
B



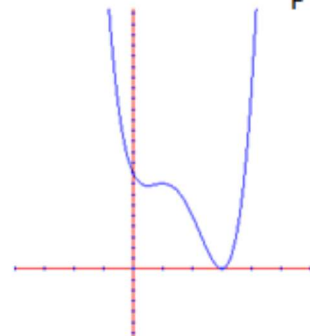
C



D



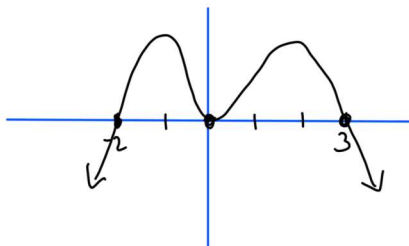
E



F

2. Write the equation in factored form of any quartic function with following characteristics. Sketch the graph of function:

- $f(0) = 0$
- $f(x) < 0$, when $x < -2$
- $f(x) \geq 0$, when $-2 \leq x \leq 3$
- $f(x) < 0$, when $x > 3$



$$f(x) = -x^2(x+2)(x-3)$$

3. Fully factor

a) $2x^5 - 2x^4 - 4x^3 + 4x^2 + 2x - 2$

$f(1) = 0 \Rightarrow x-1$ is a factor

$$\begin{array}{r|rrrrrr} 1 & 2 & -2 & -4 & 4 & 2 & -2 \\ & & 2 & 0 & -4 & 0 & 2 \\ \hline & 2 & 0 & -4 & 0 & 2 & 0 \end{array}$$

$$\begin{aligned} f(x) &= (x-1)(2x^4 - 4x^2 + 2) \\ &= 2(x-1)(x^4 - 2x^2 + 1) \\ &= 2(x-1)(x^2 - 1)^2 \\ &= 2(x-1)(x-1)^2(x+1)^2 \\ &= 2(x-1)^3(x+1)^2 \end{aligned}$$

b) $64y^6x^3 - 125$

$$= (4y^2 - 5)(16y^4x^3 + 20y^2x + 25)$$

c) $6(x+2)^{-5} + 2x^2(x+2)^{-4}$

$$\begin{aligned} &= 2(x+2)^{-5} [3 + x^2(x+2)] \\ &= 2(x+2)^{-5} [3 + x^3 + 2x^2] \\ &= 2(x+2)^{-5} (x^3 + 2x^2 + 3) \\ &= \frac{2(x^3 + 2x^2 + 3)}{(x+2)^5} \end{aligned}$$

d) $4x^4 - 13x^3 - 13x^2 + 28x - 6 = 0$

possible factors: $\pm 1, 2, 3, 6$
 $\pm 1, 2, 4$

$f(1) = 0 \Rightarrow x-1$ is a factor

$$\begin{array}{r|rrrrrr} 1 & 4 & -13 & -13 & 28 & -6 \\ & & 4 & -9 & -22 & 6 \\ \hline & 4 & -9 & -22 & 6 & 0 \end{array}$$

$$f(x) = (x-1)(4x^3 - 9x^2 - 22x + 6)$$

$f(\frac{1}{4}) = 0 \Rightarrow x - \frac{1}{4}$ is a factor

$$\begin{array}{r|rrrr} \frac{1}{4} & 4 & -9 & -22 & 6 \\ & & 1 & -2 & -6 \\ \hline & 4 & -8 & -24 & 0 \end{array}$$

$$\begin{aligned} \therefore f(x) &= (x-1)(x-\frac{1}{4})(4x^2 - 8x - 24) \\ &= (x-1)(4x-1)(x^2 - 2x - 6) \end{aligned}$$

4. Divide $8x^4 - 30x^2 + 6x - 3$ by $1 + x + 2x^2$ using **long division** and write the division statement.

$$\begin{array}{r} 4x^2 - 2x - 16 \\ 2x^2 + x + 1 \overline{) 8x^4 + 0x^3 - 30x^2 + 6x - 3} \\ \underline{- 8x^4 + 4x^3 + 4x^2} \\ -4x^3 - 34x^2 + 6x \\ \underline{- (-4x^3 - 2x^2 - 2x)} \\ -32x^2 + 8x - 3 \\ \underline{- (-32x^2 - 16x - 16)} \\ 24x + 13 \end{array}$$

$$\therefore \frac{8x^4 - 30x^2 + 6x - 3}{1 + x + 2x^2} = 4x^2 - 2x - 16 + \frac{24x + 13}{1 + x + 2x^2}$$

5. When $f(x) = x^4 - 4x^3 + mx^2 + nx + 1$ is divided by $x-1$, the remainder is 7. When it is divided by $x+1$, the remainder is 3. Determine the values of m and n .

$$f(1) = 7 \Rightarrow 1 - 4 + m + n + 1 = 7$$

$$m + n = 9 \text{ --- (1)}$$

$$f(-1) = 3 \Rightarrow 1 + 4 + m - n + 1 = 3$$

$$m - n = -3 \text{ --- (2)}$$

(1) + (2):

$$\begin{array}{r} m + n = 9 \\ +) m - n = -3 \\ \hline 2m = 6 \\ m = 3 \end{array}$$

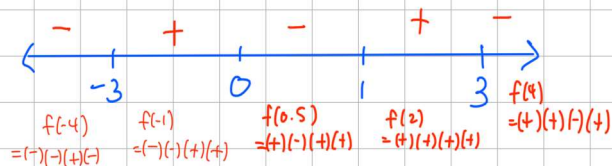
Sub. $m=3$ into (1):

$$\begin{array}{r} 3 + n = 9 \\ n = 6 \end{array}$$

6. Solve each of the following, $x \in \mathbb{R}$.

a) $x(x-1)(3-x)(x+3) < 0$

Roots: 0, 1, 3, -3



$$\therefore x \in (-\infty, -3) \cup (0, 1) \cup (3, \infty)$$

b) $x^3 - x^2 < 5x + 3$

$$x^3 - x^2 - 5x - 3 < 0$$

possible factors: $\pm 1, 3$

$$f(-1) = 0 \Rightarrow x+1 \text{ is a factor}$$

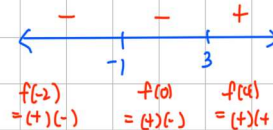
$$\begin{array}{r|rrrr} -1 & 1 & -1 & -5 & -3 \\ & & -1 & 2 & 3 \\ \hline & 1 & -2 & -3 & 0 \end{array}$$

$$(x+1)(x^2 - 2x - 3) < 0$$

$$(x+1)(x-3)(x+1) < 0$$

$$(x+1)^2(x-3) < 0$$

Roots: -1, 3



$$\therefore x \in (-\infty, -1) \cup (-1, 3)$$

7. Find the value of a and b and the remaining factor if the expression $ax^3 - 11x^2 + bx + 3$ is divisible by $x^2 - 4x + 3$.

$$x^2 - 4x + 3 = (x-3)(x-1)$$

$$(3x^3 - 11x^2 + 5x + 3) \div (x^2 - 4x + 3)$$

$$f(3) = 0 \Rightarrow 27a - 99 + 3b + 3 = 0$$

$$27a + 3b = 96$$

$$9a + b = 32 \text{ --- (1)}$$

$$f(1) = 0 \Rightarrow a - 11 + b + 3 = 0$$

$$a + b = 8 \text{ --- (2)}$$

$$\text{(1) - (2): } 9a + b = 32$$

$$-) a + b = 8$$

$$8a = 24$$

$$a = 3$$

Sub. $a=3$ into (2):

$$3 + b = 8$$

$$b = 5$$

$$\begin{array}{r} 4 \rightarrow \begin{array}{rrrr} 3 & -11 & 5 & 3 \\ & -12 & -4 & -3 \\ \hline & 1 & 9 & 0 \\ & & -3 & -5 \\ \hline & & 0 & 0 \end{array} \\ \downarrow \\ 3x + 1 \end{array}$$

$\therefore$ The remaining factor is $3x+1$

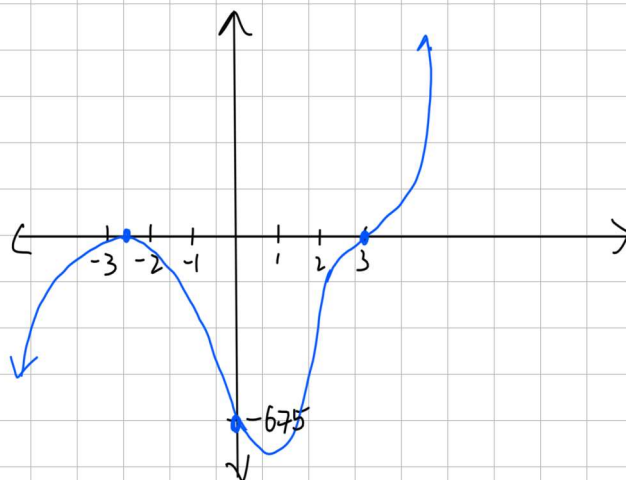
8. Graph the following function $f(x) = (x^2+x+1)(2x+5)^2(x-3)^3$. Show all your work.

Roots: $-\frac{5}{2}$ (double)
3 (triple)

degree: 7

leading coefficient: +2

y-intercept: $y = (1)(5)^2(-3)^3$
 $= -675$



9. The passenger section of a train has a width $2x - 7$, length $2x + 3$, and height $x - 2$, with all dimensions in metres. Solve a polynomial equation to determine the dimensions of the section of the train if the volume is 117m^3 .

$$(2x-7)(2x+3)(x-2) = 117$$

$$4x^3 - 16x^2 - 5x + 42 = 117$$

$$4x^3 - 16x^2 - 5x - 75 = 0$$

possible factors: $\frac{\pm 1, 3, 5, 15, 25, 75}{\pm 1, 2, 4}$

$f(5) = 0 \Rightarrow x-5$ is a factor

5	4	-16	-5	-75
		20	20	75
4	4	15	0	

$$(x-5)(4x^2+4x+15) = 0$$

$\downarrow$
 $x=5$

no real solutions

$$2x-7 = 2(5)-7 = 3$$

$$2x+3 = 2(5)+3 = 13$$

$$x-2 = 5-2 = 3$$

$\therefore$ The dimensions are 3 m by
13 m by 3 m

10. Determine algebraically, whether each function is even, odd, or neither.

a) $f(x) = 4x^3$

$$\begin{aligned} f(-x) &= 4(-x)^3 \\ &= -4x^3 \\ &= -f(x) \end{aligned}$$

$\therefore$ Odd

b) $f(x) = 2x^4 - x^2$

$$\begin{aligned} f(-x) &= 2(-x)^4 - (-x)^2 \\ &= 2x^4 - x^2 \\ &= f(x) \end{aligned}$$

$\therefore$ Even

c) $g(x) = \sqrt[3]{2x^2 + 1}$

$$\begin{aligned} g(-x) &= \sqrt[3]{2(-x)^2 + 1} \\ &= \sqrt[3]{2x^2 + 1} \\ &= g(x) \end{aligned}$$

$\therefore$ Even

d) $h(x) = \frac{-x^3}{(3x^3 - 9x)^2}$

$$\begin{aligned} h(-x) &= \frac{-(-x)^3}{[3(-x)^3 - 9(-x)]^2} \\ &= \frac{x^3}{(-3x^3 + 9x)^2} \\ &= \frac{x^3}{[-(3x^3 - 9x)]^2} \\ &= \frac{x^3}{(3x^3 - 9x)^2} \\ &= -f(x) \end{aligned}$$

$\therefore$ Odd

e) $f(x) = x + |x|$

$$\begin{aligned} f(-x) &= -x + |-x| \\ &= -x + |x| \\ &\neq -f(x) \neq f(x) \end{aligned}$$

$\therefore$ Neither

f) $g(x) = \frac{2x}{|x|}$

$$\begin{aligned} g(-x) &= \frac{2(-x)}{|-x|} \\ &= \frac{-2x}{|x|} \\ &= -g(x) \end{aligned}$$

$\therefore$ Odd

11. The table of values below represents a polynomial function. Determine the equation of this function.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
-2	-19			
-1	-3	16		
0	1	4	-12	
1	-1	-2	-6	6
2	-3	-2	0	6
3	1	4	6	6

$$6 = a \cdot 3!$$

$$6 = 6a$$

$$1 = a$$

$$y = ax^3 + bx^2 + cx + d$$

$$y = x^3 + bx^2 + cx + d$$

Sub. (0,1): $1 = d$

Sub. (1,-1): $-1 = 1 + b + c + 1$
 $-3 = b + c$ — (1)

Sub. (-1,-3): $-3 = -1 + b - c + 1$
 $-3 = b - c$ — (2)

(1)+(2): $-3 = b + c$
 $+ \quad -3 = b - c$
 $\hline -6 = 2b$
 $-3 = b$

Sub. $b = -3$ into (1):
 $-3 = -3 + c$
 $0 = c$

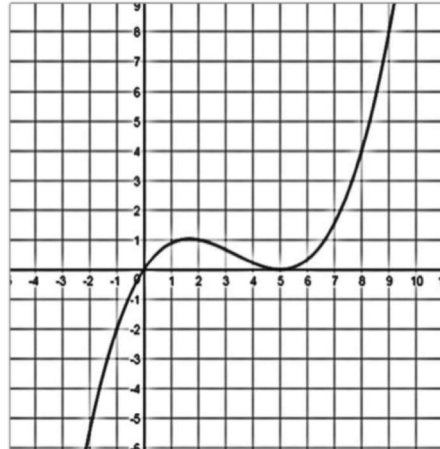
$$\therefore y = x^3 - 3x^2 + 1$$

12. Find the general equation of quartic functions that has negative leading coefficient, two equal roots at 2, and roots at $3 \pm 2\sqrt{2}$.

$$\begin{aligned}
 x &= 3 \pm 2\sqrt{2} \\
 (x-3)^2 &= (\pm 2\sqrt{2})^2 \\
 x^2 - 6x + 9 &= 4(2) \\
 x^2 - 6x + 9 &= 8 \\
 x^2 - 6x + 1 &= 0
 \end{aligned}
 \quad \therefore f(x) = k(x-2)^2(x^2-6x+1), \quad k < 0$$

13. Given the graph of a polynomial function $g(x)$, answer the following:

- Is the function even-degree or odd-degree? Odd
- Is the function even or odd or neither? Neither
- State the zeroes and the lowest possible order of each zero 0 (order 1), 5 (order 2)
- State the interval where the function is positive $x \in (0, 5) \cup (5, \infty)$
- State the interval where the function is negative $x \in (-\infty, 0)$
- Determine the value of the remainder when $f(x)$ is divided by $x+2$. $f(-2) = -5$



14. Water is draining from a container. The height, in millimeters, of the water as a function of time, in seconds, can be modeled by the function

$$h(t) = 0.00185(250 - t)^2.$$

- a) Calculate the average rate of change of height with respect to time from 50s to 100s.

$$A.R.O.C = \frac{h(100) - h(50)}{100 - 50} = \frac{41.625 - 74}{50} = -0.6475 \text{ mL/s}$$

- b) Calculate the instantaneous rate of change of height with respect to time at $t=60$ s.

$$I.R.O.C = \frac{h(60.001) - h(60)}{0.001} = \frac{66.784297 - 66.785}{0.001} = -0.703 \text{ mL/s}$$

- c) Create a sketch of the function indicating the secant line and tangent line from part a.



15. When polynomial $x^3 - ax + 21$ is divided by $x + b$, the quotient is $x^2 - 3x + 5$ and the remainder is 6. Determine values of **a** and **b**.

$$\begin{aligned} f(x) = x^3 - ax + 21 &= (x^2 - 3x + 5)(x + b) + 6 \\ &= x^3 + bx^2 - 3x^2 - 3xb + 5x + 5b + 6 \\ &= x^3 + (b-3)x^2 + (-3b+5)x + (5b+6) \end{aligned}$$

$$0 = b - 3$$

$$3 = b$$

$$-a = -3b + 5$$

$$-a = -9 + 5$$

$$a = 4$$

16. Is $x+b$ a factor of $x^9 + 5b^2x^7 + 5bx^8 - b^9$?

$$\begin{aligned} f(-b) &= (-b)^9 + 5b^2(-b)^7 + 5b(-b)^8 - b^9 \\ &= -b^9 - 5b^9 + 5b^9 - b^9 \\ &= -2b^9 \\ &\neq 0 \end{aligned}$$

$\therefore x+b$ is not a factor.